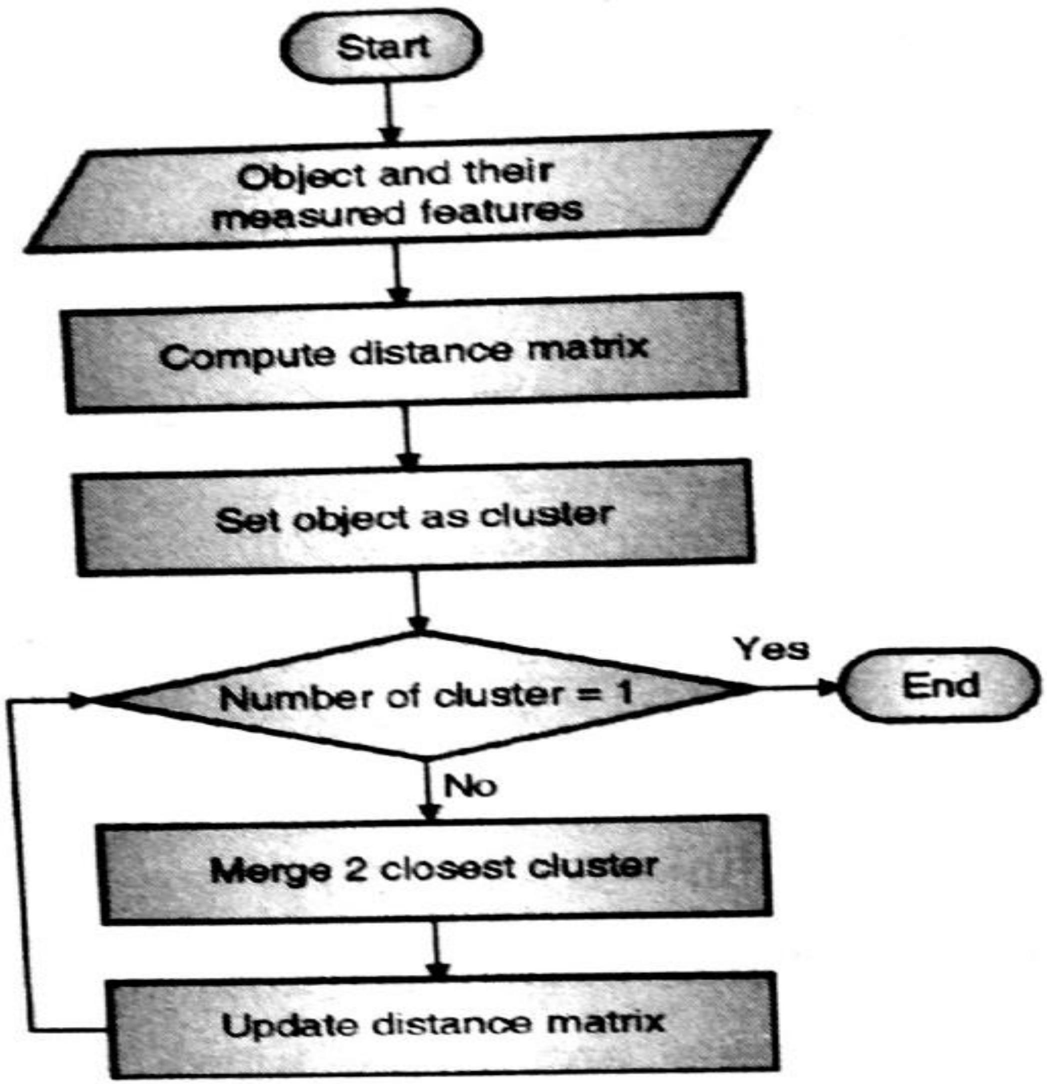
Flowchart of agglomerative hierarchical clustering

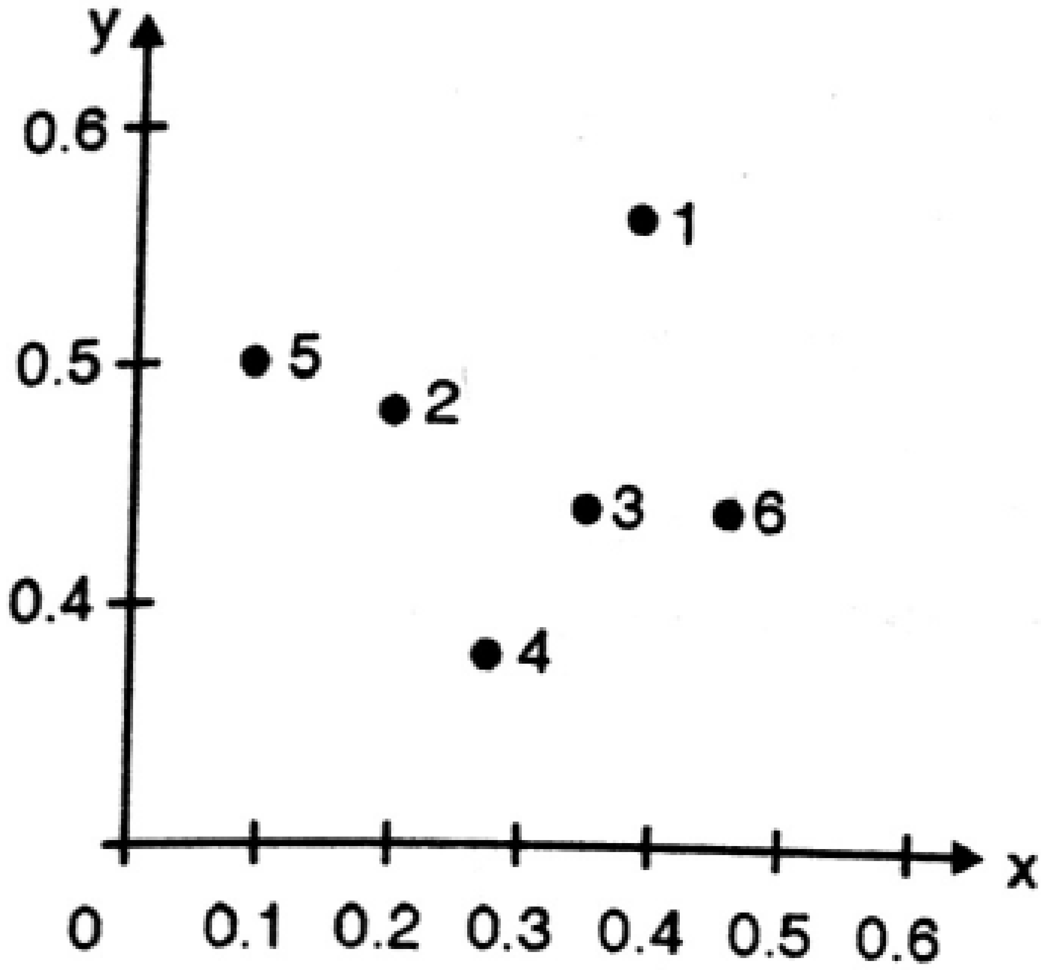


Single Link Technique

Single Link Technique

Points	X	Y
P1	0.40	0.53
p2	0.22	0.38
p3	0.35	0.32
p4	0.26	0.19
p5	0.08	0.41
p6	0.45	0.30

Plot the objects in *n*-dimensional space



Distance Matrix using Euclidean Distance

$$
\begin{aligned}
d(p1, p2) &= \sqrt{|xp_1 - xp_1|^2 + |yp_1 - yp_2|^2} \\
&= \sqrt{|0.40 - 0.22|^2 + |0.53 - 0.38|^2} \\
&= \sqrt{|0.18|^2 + |0.15|^2} \\
&= \sqrt{0.0324 + 0.0225} \\
&= \sqrt{0.0549} \\
&= 0.2343
\end{aligned}
$$

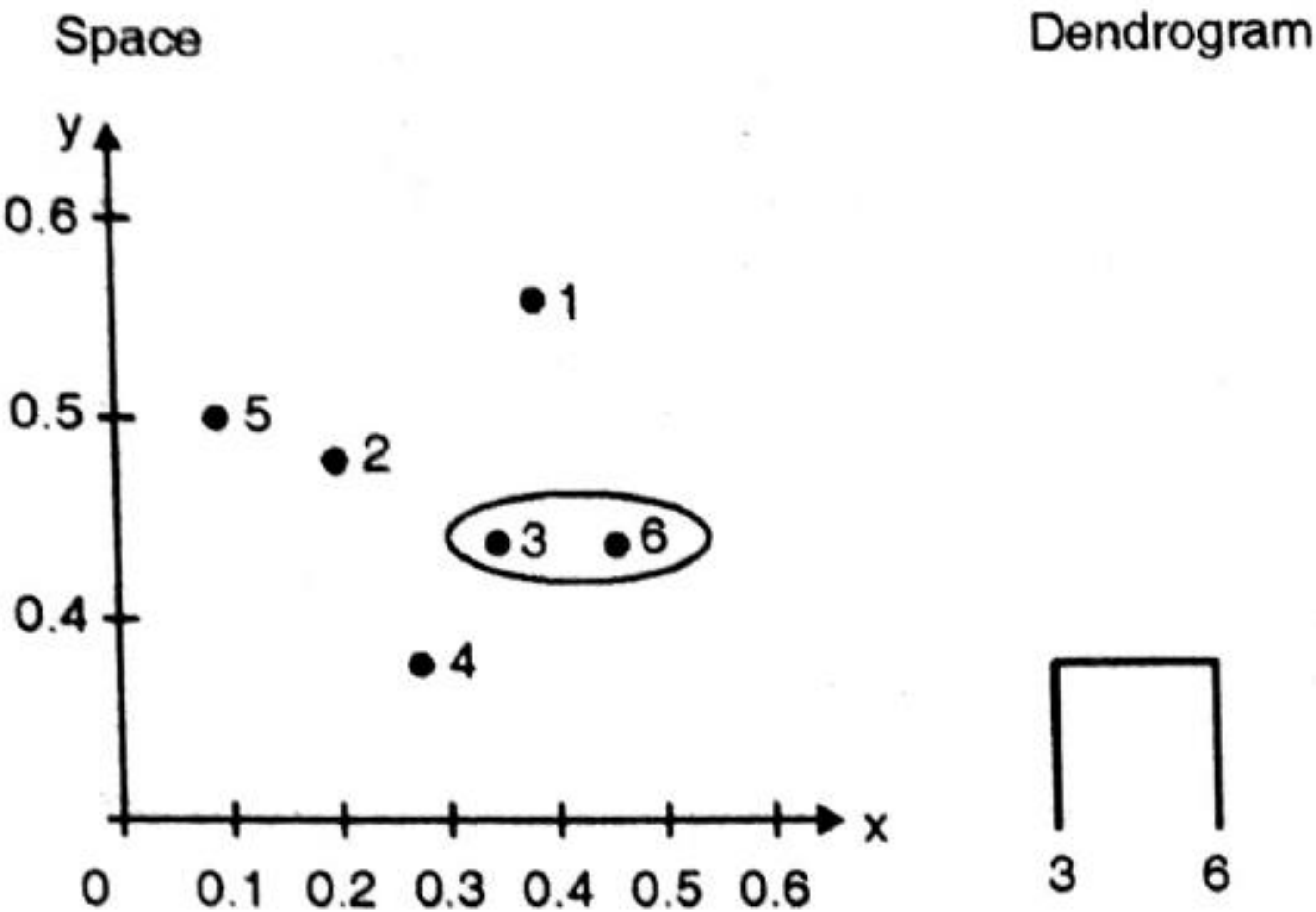
Distance Matrix using Euclidean Distance

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.24	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	**0.11**	0.22	0.39	0

Distance Matrix using Euclidean Distance

- Identify the two clusters with the shortest distance in the matrix and merge them together.
- P3 and P6 have the smallest value of all - 0.11
- P3 and P6 will be clustered.
- Re-compute the distance matrix.

Space Dendrogram



Re-compute the distance matrix

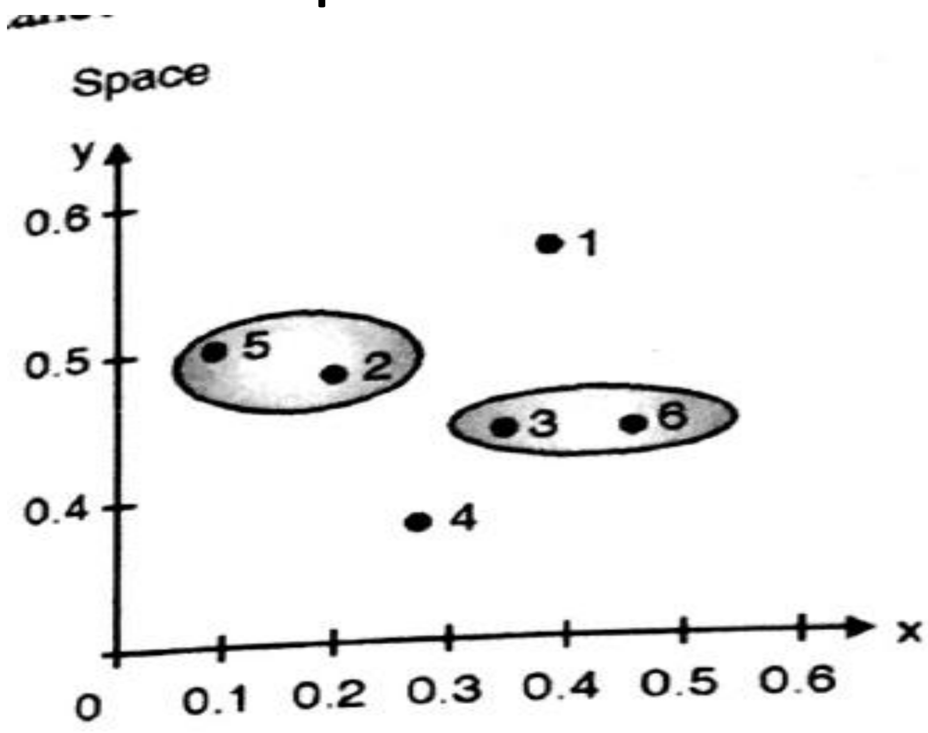
- Dist((p3,p6) , p1) = Min(dist(p3,p1) , dist(p6,p1))
  = Min(0.22,0.23)
  = 0.22
- Dist((p3,p6) , p2) = Min (dist(p3,p2) , (p6,p2))
  = Min (0.15 ,0.25)
  = 0.15

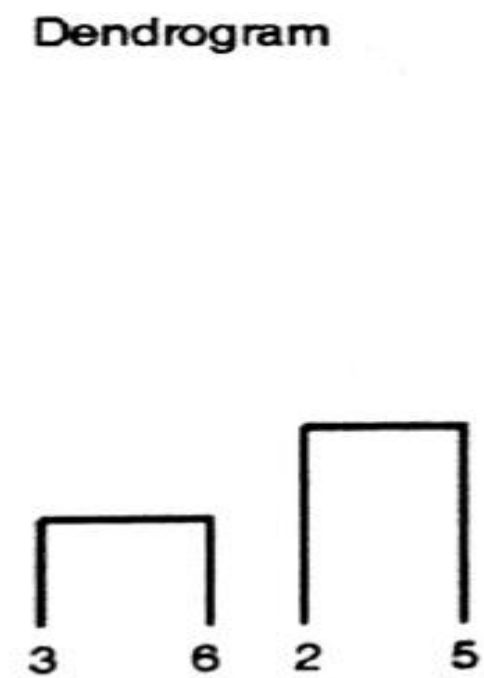
Distance Matrix

	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.24	0			
P3,P6	**0.22**	**0.15**	0		
P4	0.37	0.20	**0.15**	0	
P5	0.34	**0.14**	**0.28**	0.29	0

Space Dendrogram

- P5 and P2 have the smallest value of all - 0.14
- P5 and P2 will be clustered.
- Re-compute the distance matrix.





Re-compute the distance matrix

- dist(((P3,p6),(p2,p5)) = MIN (dist (p3,p2), dist (p6,p2), dist(p3,p5),dist (p6,p5))

  = MIN (0.15,0.25,0.28,.39)

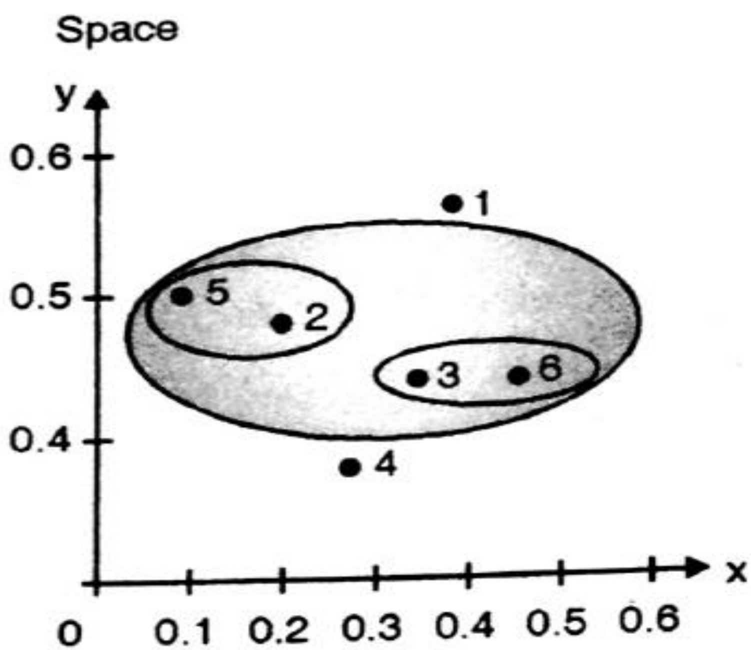
  = 0.15

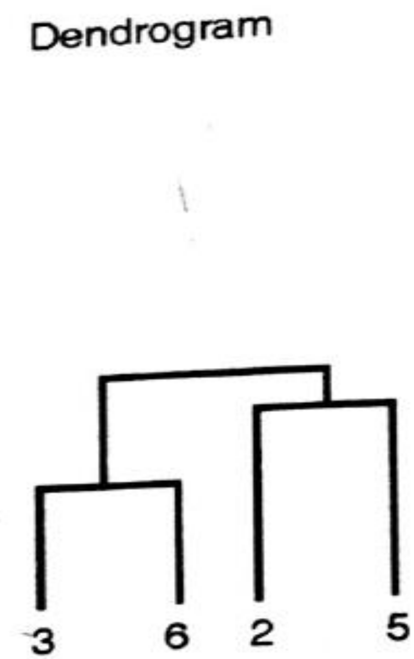
Distance Matrix

	P1	P2,P5	P3,P6	P4
P1	0			
P2,P5	**0.24**	0		
P3,P6	0.22	**0.15**	0	
p4	0.37	**0.20**	0.15	0

Space Dendrogram

- (P5,P2) and (P3,P6) have the smallest value of all - 0.15
- P4 and (P3,P6) will also have the same value.
- Choose either one and merge.
- Re-compute the distance matrix.





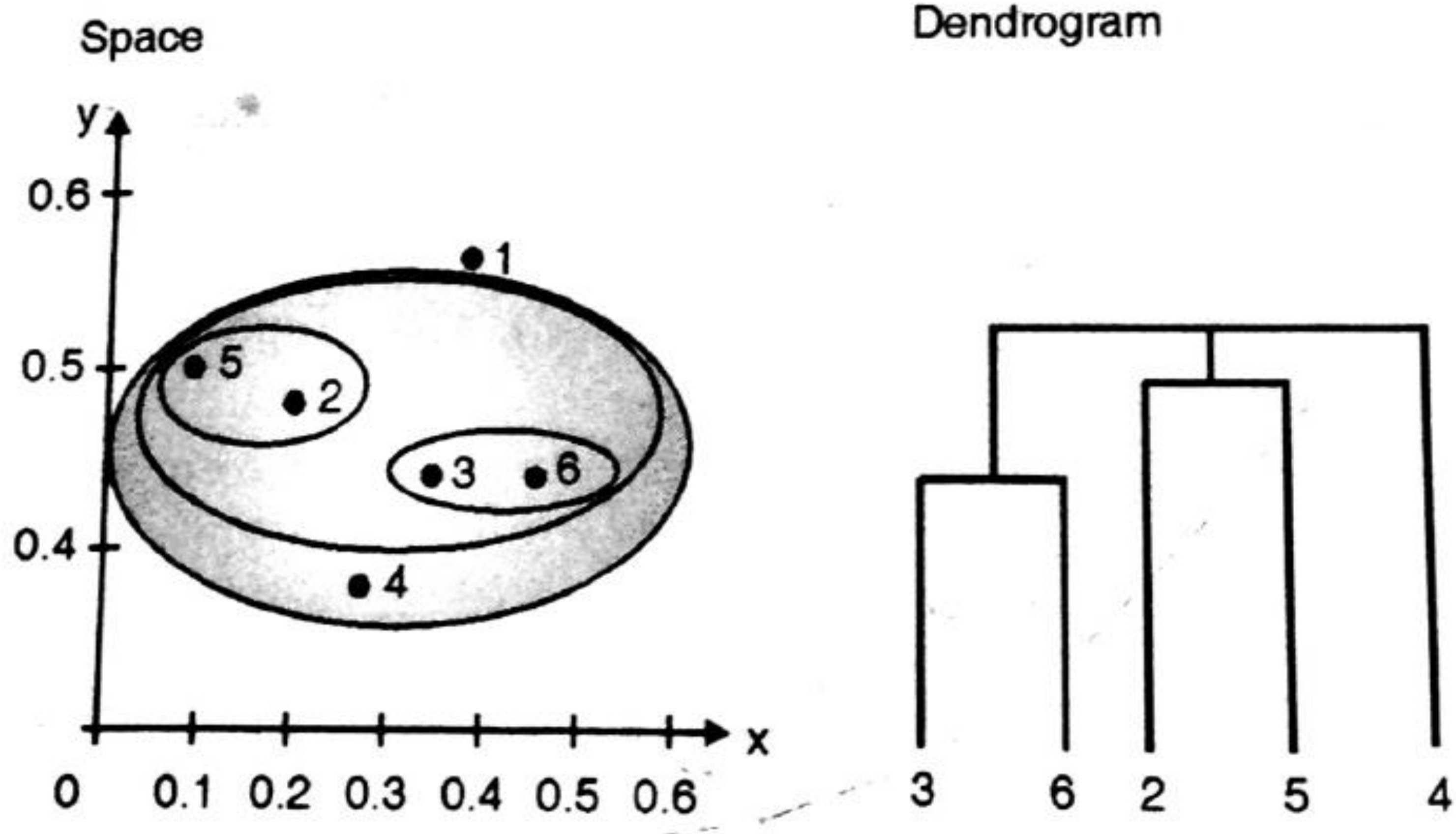
Distance Matrix

	P1	P2,P5,P3,P6	P4
P1	0		
P2,P5,P3,P6	**0.22**	0	
P4	0.37	**0.15**	0

- (P2,P5,P3,P6) and P4 have the smallest value of all - 0.15
- Merge those two in a single cluster.
- Re-compute the distance matrix.

Space Dendrogram

Space dendrogram :

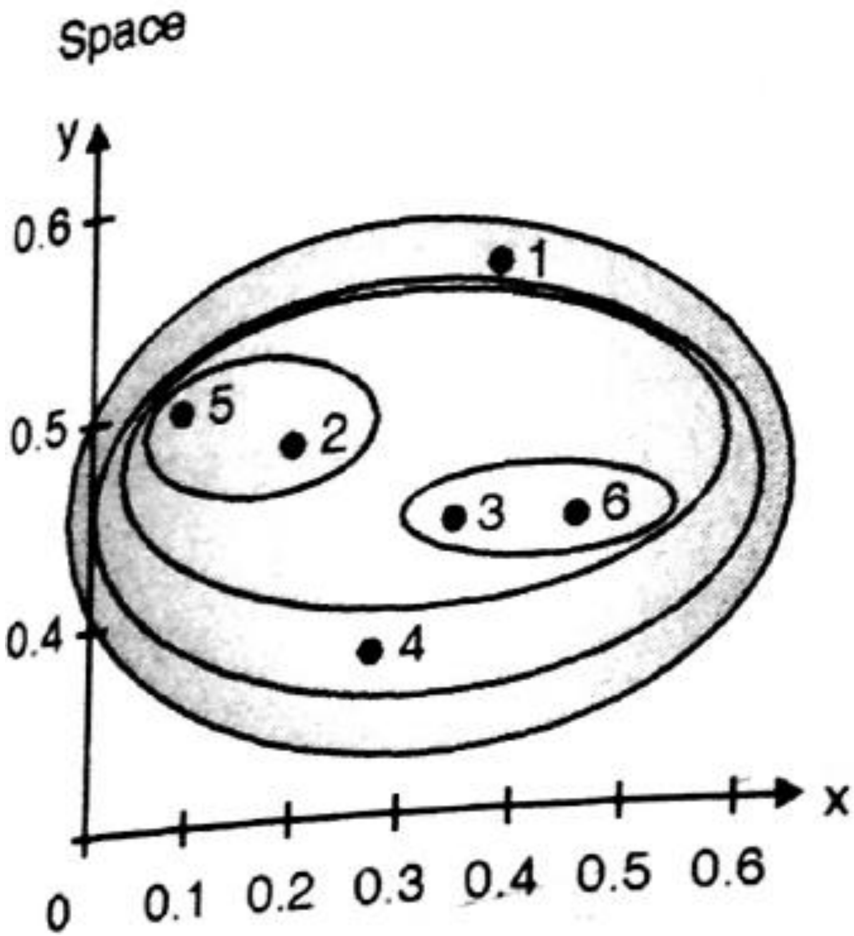


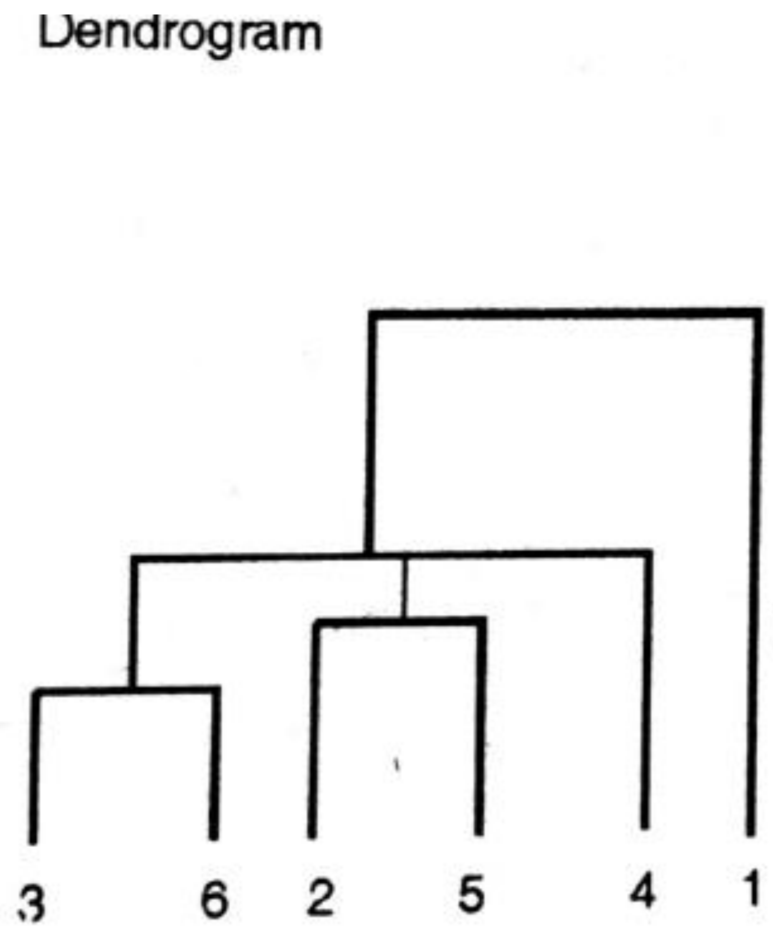
Distance Matrix

	P1	P2,P5,P3,P6,P4
P1	0	
P2,P5,P3,P6,P4	**0.22**	0

- (P2,P5,P3,P6,P4) and P1 have the smallest value of all - 0.22
- Merge those two in a single cluster.
- No need to Re-compute the distance matrix as there are no more cluster to merge.

Space Dendrogram



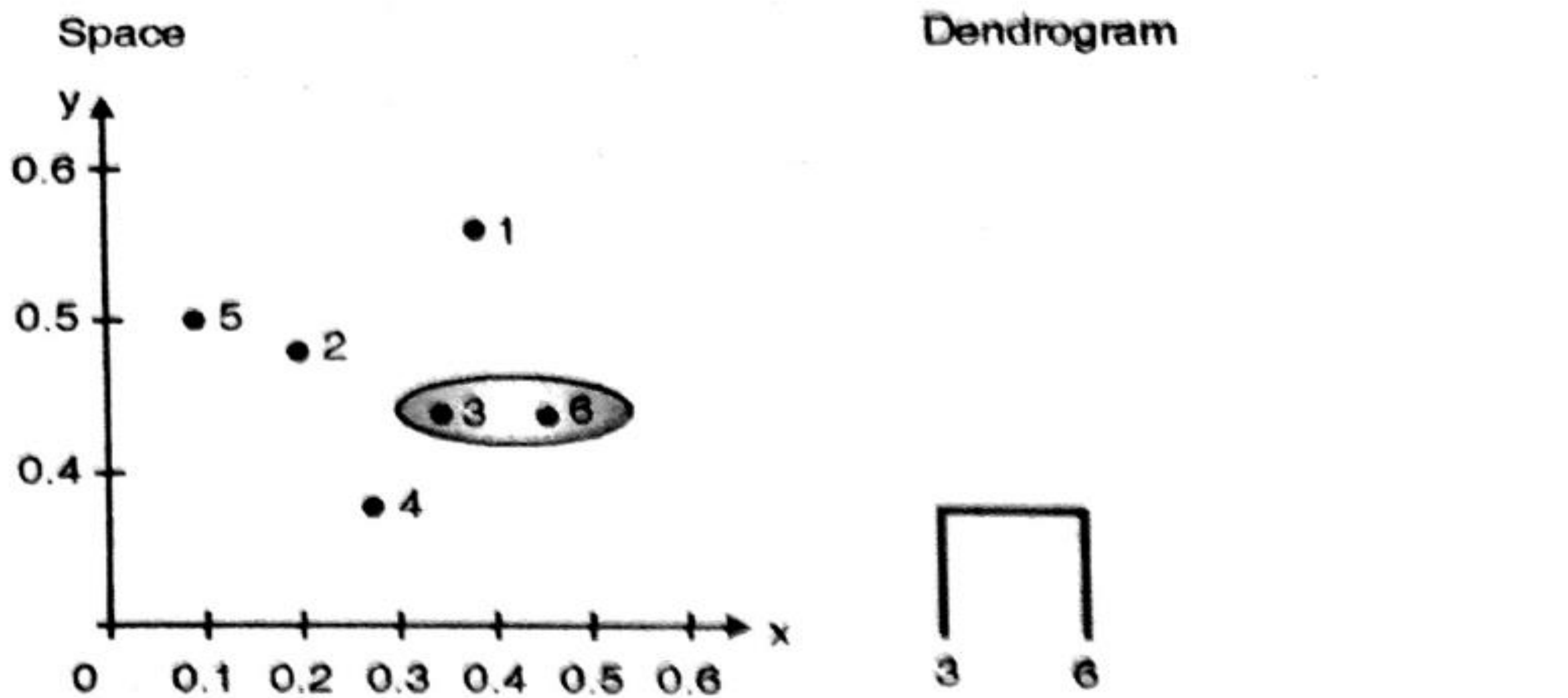


Complete Link Technique

p1	0					
p2	0.24	0				
p3	0.22	0.15	0			
p4	0.37	0.20	0.15	0		
p5	0.34	0.14	0.28	0.29	0	
p6	0.23	0.25	0.11	0.22	0.39	0
	p1	p2	p3	p4	p5	p6

Identify the two clusters with the shortest distance in the matrix, and merge them together. Re-compute the distance matrix, as those two clusters are now in a single cluster.

By looking at the distance matrix above, we see that p3 and p6 have the smallest distance from all i.e. 0.11 So, we merge those two in a single cluster, and re-compute the distance matrix.



$dist((p3, p6), p1) = MAX(dist(p3, p1), dist(p6, p1))$

$= MAX(0.22, 0.23)$ //from original matrix

$= 0.23$

$dist((p3, p6), p2) = MAX(dist(p3, p2), dist(p6, p2))$

$= MAX(0.15, 0.25)$ //from original matrix

$= 0.25$

$dist((p3, p6), p4) = MAX(dist(p3, p4), dist(p6, p4))$

$= MAX(0.15, 0.22)$ //from original matrix

$= 0.22$

$dist((p3, p6), p5) = MAX(dist(p3, p5), dist(p6, p5))$

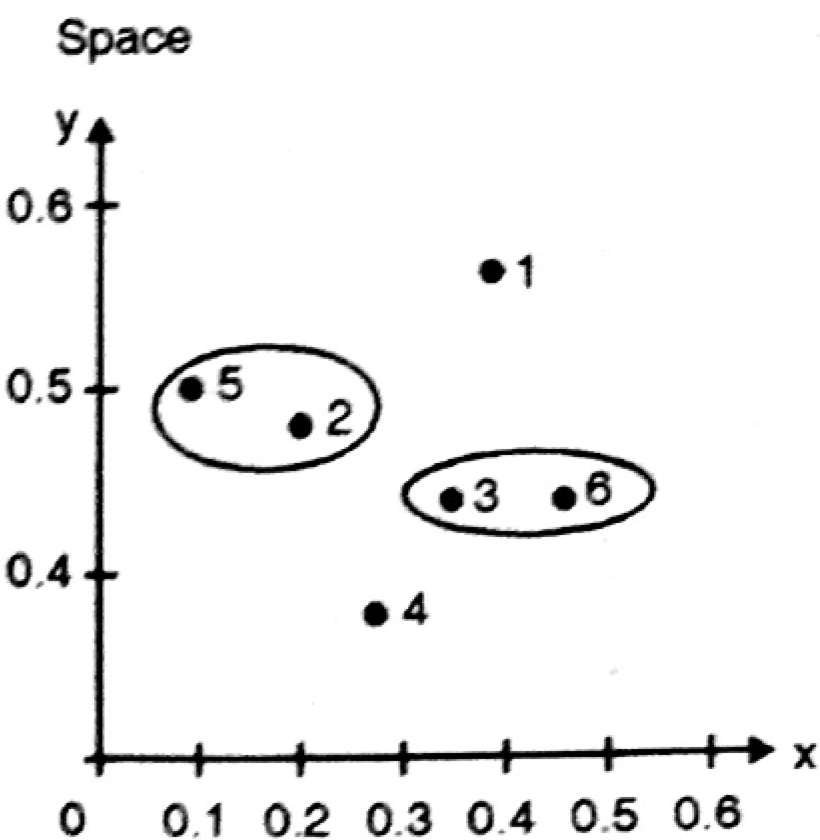
$= MAX(0.28, 0.39)$ //from original matrix

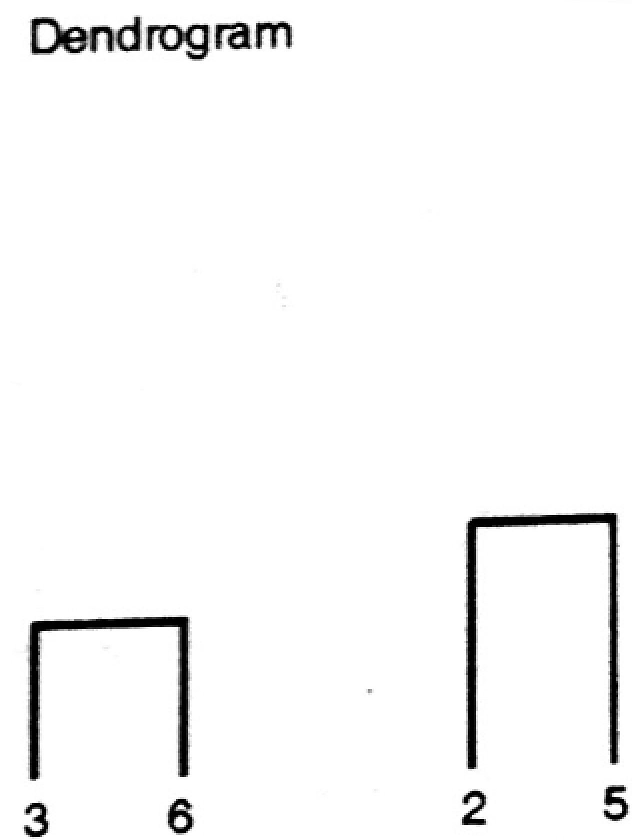
$= 0.39$

New Distance matrix :

p1	0				
p2	0.24	0			
(p3, p6)	**0.23**	**0.25**	0		
p4	0.37	0.20	**0.22**	0	
p5	0.34	0.14	**0.39**	0.29	0
	p1	p2	(p3, p6)	p4	p5

So, looking at the above distance matrix, we see that p2 and p5 have the smallest distance from all - 0.14. So, we merge those two in a single cluster, and re-compute the distance matrix using the following calculations





$$dist((p2, p5), p1) = MAX(dist(p2, p1), dist(p5, p1))$$

$$= MAX(0.24, 0.34) \quad \text{//from original matrix}$$

$$= 0.34$$

$$dist((p2, p5), (p3,p6)) = MAX(dist(p2, p3), dist(p2, p6), dist(p5, p3), dist(p5, p6))$$

$$= MAX(0.15, 0.25, 0.28, 0.39) \quad \text{//from original matrix}$$

$$= 0.39$$

$$dist((p2, p5), p4) = MAX(dist(p2, p4), dist(p5, p4))$$

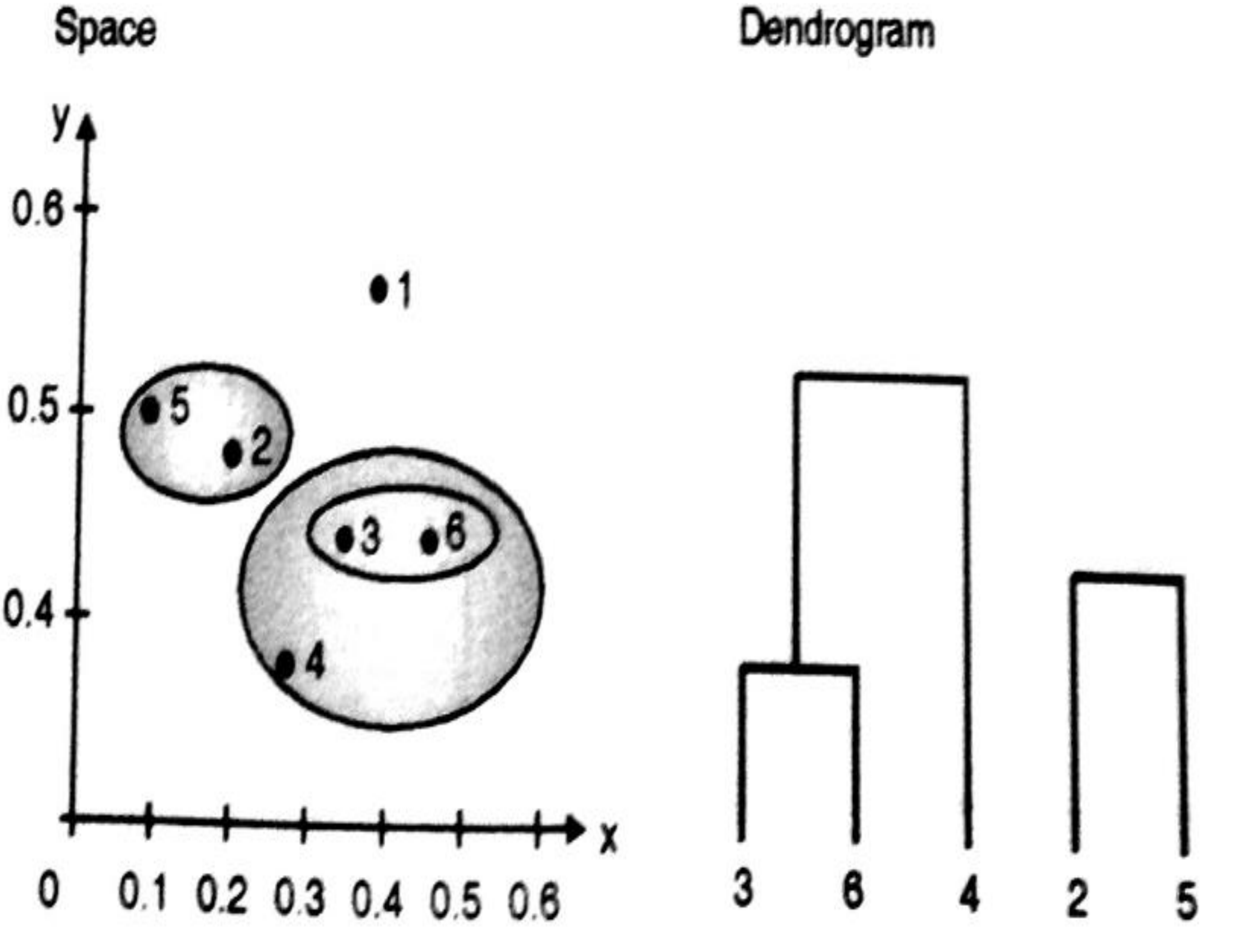
$$= MAX(0.20, 0.29) \quad \text{//from original matrix}$$

$$= 0.29$$

Therefore new Distance matrix is :

p1	0			
(p2, p5)	**0.34**	0		
(p3, p6)	0.23	**0.39**	0	
p4	0.37	**0.29**	0.22	0
	p1	(p2, p5)	(p3, p6)	p4

The minimum distance is 0.22, so merge (p3, p6) and p4



Space
Dendrogram
y
0.6
0.5
0.4
x
0
0.1
0.2
0.3
0.4
0.5
0.6
1
5
2
3
6
4
3
6
4
2
5

$$dist((p3, p6, p4), p1) = \text{MAX}(dist(p3, p1), dist(p6, p1), dist(p4, p1))$$

$$= \text{MAX}(0.22, 0.23, 0.37) \quad \text{//from original matrix}$$

$$= 0.37$$

$$dist((p3, p6, p4), (p2,p5)) = \text{MAX}(dist(p3, p2), dist(p3, p5), dist(p6, p2), dist(p6, p5), dist(p4, p2), dist(p4, p5))$$

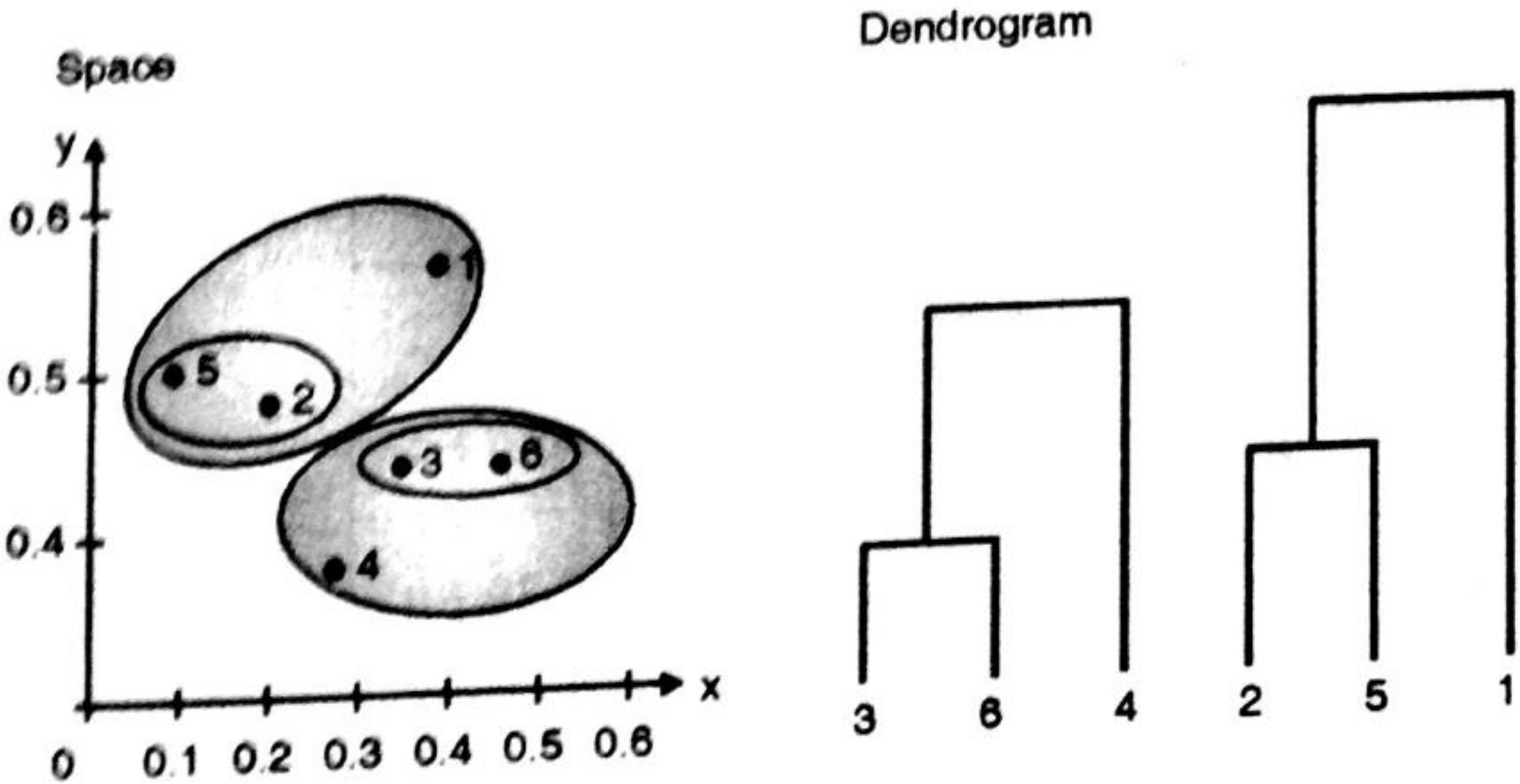
$$= \text{MAX}(0.15, 0.28, 0.25, 0.39, 0.20, 0.29)$$

$$= 0.39$$

Therefore New Distance matrix

p1	0		
(p2, p5)	**0.34**	0	
(p3, p6, p4)	0.37	**0.39**	0
	p1	**(p2, p5)**	(p3, p6, P4)

Since the minimum distance is 0.34, merge (p2, p5) with p1

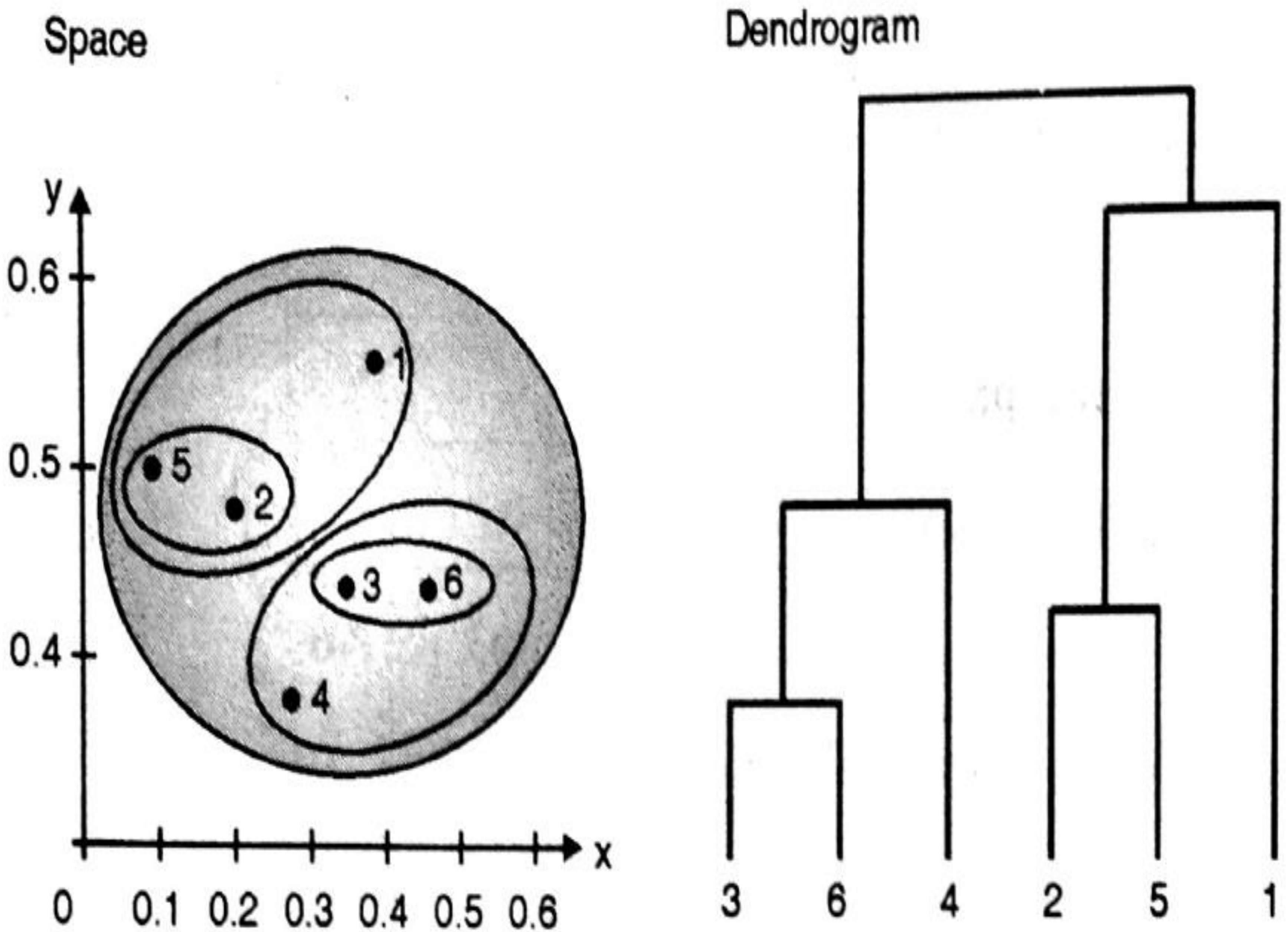


Dist ((p2, p5, p1), (p3, p6, p4) = 0.39

Therefore New Distance matrix

(p2, p5,p1)	0	
(p3, p6, p4)	0.39	0
	(p2, p5,p1)	(p3, p6, P4)

Finally, Merge the cluster (p2, p5,p1) and (p3,p6,p4)


Space
y
0.6
0.5
0.4
x
0
0.1
0.2
0.3
0.4
0.5
0.6
1
5
2
3
6
4
Dendrogram
3
6
4
2
5
1

Average Link Technique

By looking at the distance matrix above, we see that p3 and p6 have the smallest distance from all – 0.11 So, we merge those two in a single cluster, and re-compute the distance matrix.

$$\text{dist}((p3\ p6),\ p1) = 1/2\,(\text{dist}(p3,p1) + \text{dist}(p6,p1))$$
$$= 0.5 \times (0.22 + 0.23) = 0.23$$

$$\text{dist}((p3\ p6),\ p2) = 1/2\,(\text{dist}(p3,p2) + \text{dist}(p6,p2))$$
$$= 0.5 \times (0.15 + 0.25) = 0.2$$

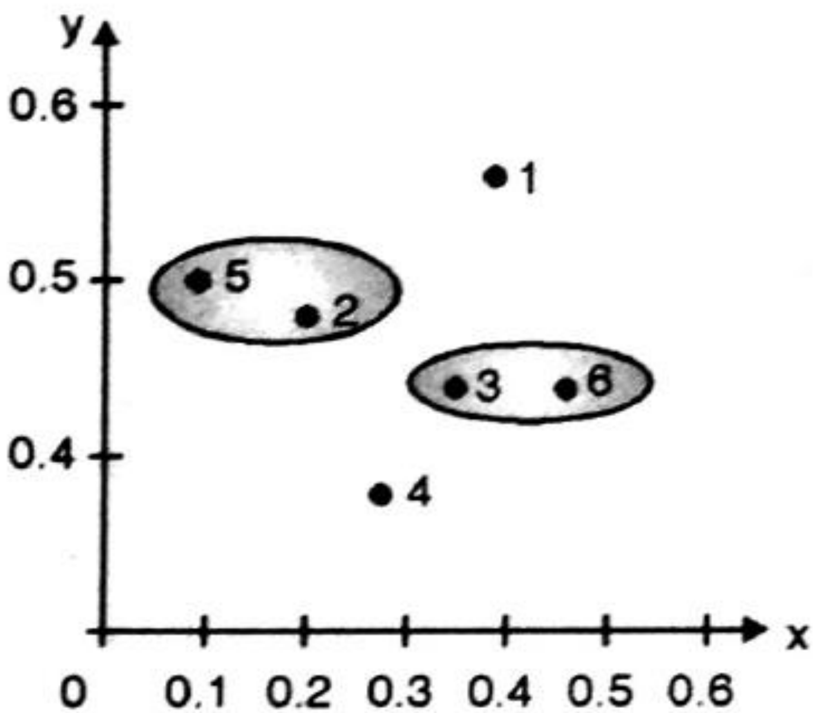
$$\text{dist}((p3\ p6),\ p4) = 1/2\,(\text{dist}(p3,p4) + \text{dist}(p6,p4))$$
$$= 0.5 \times (0.15 + 0.22) = 0.19$$

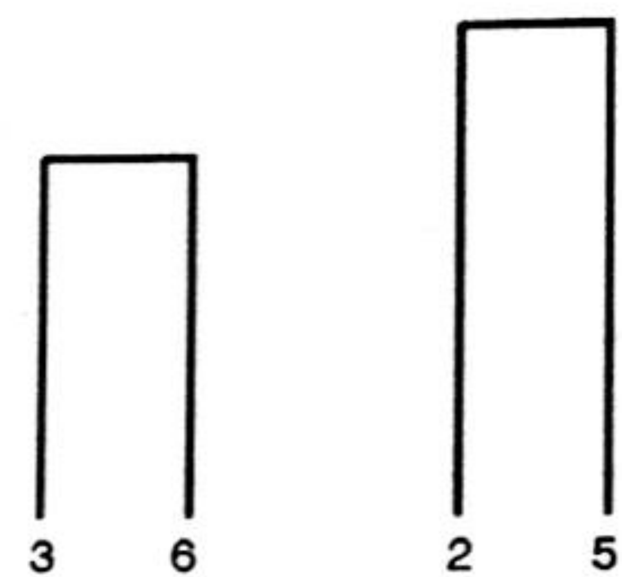
$$\text{dist}((p3\ p6),\ p5) = 1/2\,(\text{dist}(p3,p5) + \text{dist}(p6,p5))$$
$$= 0.5 \times (0.28 + 0.39) = 0.34$$

Distance matrix

	p1	p2	(p3, p6)	p4	p5
p1	0				
p2	0.24	0			
(p3, p6)	**0.23**	**0.2**	0		
p4	0.37	0.20	**0.19**	0	
p5	0.34	0.14	**0.34**	0.29	0

So, looking at the above distance matrix above, we see that p2 and p5 have the smallest distance from all – 0.14 So, we merge those two in a single cluster, and re-compute the distance matrix.





$$
\begin{aligned}
\text{dist}((p2\ p5),\ p1) &= 1/2\,(\text{dist}(p2,p1) + \text{dist}(p5,p1)) \\
&= 0.5 \times (0.24 + 0.34) \\
&= 0.29
\end{aligned}
$$

$$
\begin{aligned}
\text{dist}((p2\ p5),\ (p3,p6)) &= 1/4\,(\text{dist}(p2,p3) + \text{dist}(p2,p6) + \text{dist}(p5,p3) + \text{dist}(p5,p6)) \\
&= 1/4 \times (0.15 + 0.25 + 0.28 + 0.39) \\
&= 0.27
\end{aligned}
$$

$$
\begin{aligned}
\text{dist}((p2\ p5),\ p4) &= 1/2\,(\text{dist}(p2,p4) + \text{dist}(p5,p4)) \\
&= 0.5 \times (0.14 + 0.29) \\
&= 0.22
\end{aligned}
$$

Distance matrix

p1	0			
(p2, p5)	**0.29**	0		
(p3, p6)	0.22	**0.27**	0	
p4	0.37	**0.22**	0.15	0
	p1	(p2, p5)	(p3, p6)	p4

- Minimum dist is 0.15.
- Merge P3,P6 and P4.
- Re-compute the distance matrix.

$$\begin{aligned}
\text{dist}((p3\ p6\ p4), (p2, p5)) &= 1/6 \times (\text{dist}(p3,p2) + \text{dist}(p3,p5) + \text{dist}(p6,p2) \\
&\quad + \text{dist}(p6,p5) + \text{dist}(p4,p2) + \text{dist}(p4,p5)) \\
&= 1/6 \times (0.15 + 0.28 + 0.25 + 0.39 + 0.20 + 0.29) \\
&= 0.26
\end{aligned}$$

$$\begin{aligned}
\text{dist}((p3\ p6\ p4), p1) &= 1/3 \times (\text{dist}(p3,p1) + \text{dist}(p6,p1) + \text{dist}(p4,p1)) \\
&= 1/3 \times (0.22 + 0.23 + 0.37) \\
&= 0.27
\end{aligned}$$

	P1	P2,P5	P3,P6,P4
P1	0		
P2,P5	0.29	0	
P3,P6,P4	**0.27**	**0.26**	0

- Minimum dist is 0.26.
- Merge (P3,P6,P4) and (P2,P5).
- Re-compute the distance matrix.

dist(((P3,p6,P4,p2,p5),(P1) = 1/5 (dist (p3,p1) + dist (p6,p1) + dist (p4,p1) + dist (p2,p1) + dist(p5,p1))

=1/5 (0.22 + 0.23 + 0.37 + 0.24 + 0.34)

= 0.28

	P1	P3,P6,P4,P2,P5
P1	0	
P3,P6,P4,P2,P5	0.26	0

For given distance matrix, draw single link, complete link and average link dendrogram.

$$\begin{array}{c c}
 & \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\
\begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} & \left[\begin{array}{ccccc} 0 & & & & \\ 2 & 0 & & & \\ 6 & 3 & 0 & & \\ 10 & 9 & 7 & 0 & \\ 9 & 8 & 5 & 4 & 0 \end{array}\right]
\end{array}$$

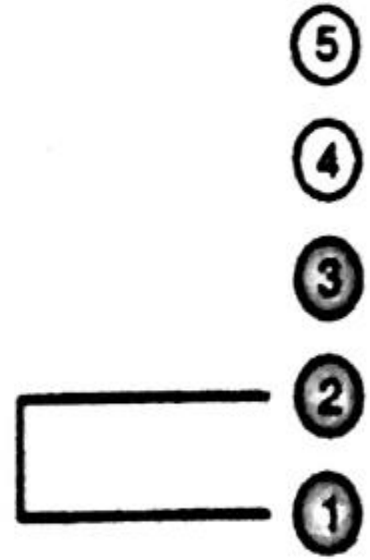
Step 1 :

$$
\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array}
\begin{array}{c}
\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\
\begin{bmatrix}
0 & & & & \\
2 & 0 & & & \\
6 & 3 & 0 & & \\
10 & 9 & 7 & 0 & \\
9 & 8 & 5 & 4 & 0
\end{bmatrix}
\end{array}
\Rightarrow
\begin{array}{c} \\ (1,2) \\ 3 \\ 4 \\ 5 \end{array}
\begin{array}{c}
\begin{array}{cccc} (1,2) & 3 & 4 & 5 \end{array} \\
\begin{bmatrix}
0 & & & \\
3 & 0 & & \\
9 & 7 & 0 & \\
8 & 5 & 4 & 0
\end{bmatrix}
\end{array}
$$

$$d_{(1,2)3} = \min\{d_{1,3}, d_{2,3}\} = \min\{6, 3\} = 3$$

$$d_{(1,2)4} = \min\{d_{1,4}, d_{2,4}\} = \min\{10, 9\} = 9$$

$$d_{(1,2)5} = \min\{d_{1,5}, d_{2,5}\} = \min\{9, 8\} = 8$$

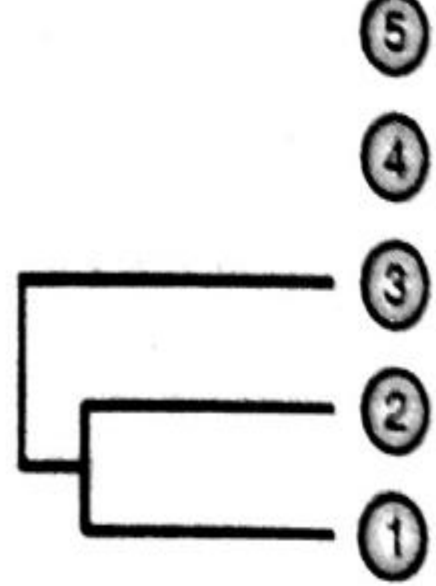


Step 2 :

$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} \begin{array}{c} \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\ \begin{bmatrix} 0 & & & & \\ 2 & 0 & & & \\ 6 & 3 & 0 & & \\ 10 & 9 & 7 & 0 & \\ 9 & 8 & 5 & 4 & 0 \end{bmatrix} \end{array} \Rightarrow \begin{array}{c} \\ (1,2) \\ 3 \\ 4 \\ 5 \end{array} \begin{array}{c} \begin{array}{cccc} (1,2) & 3 & 4 & 5 \end{array} \\ \begin{bmatrix} 0 & & & \\ 3 & 0 & & \\ 9 & 7 & 0 & \\ 8 & 5 & 4 & 0 \end{bmatrix} \end{array} \Rightarrow \begin{array}{c} \\ (1,2,3) \\ 4 \\ 5 \end{array} \begin{array}{c} \begin{array}{ccc} (1,2,3) & 4 & 5 \end{array} \\ \begin{bmatrix} 0 & & \\ 7 & 0 & \\ 5 & 4 & 0 \end{bmatrix} \end{array}$$

$$d_{(1,2,3),4} = \min\{d_{(1,2),4}, d_{3,4}\} = \min\{9, 7\} = 7$$

$$d_{(1,2,3),5} = \min\{d_{(1,2),5}, d_{3,5}\} = \min\{8, 5\} = 5$$



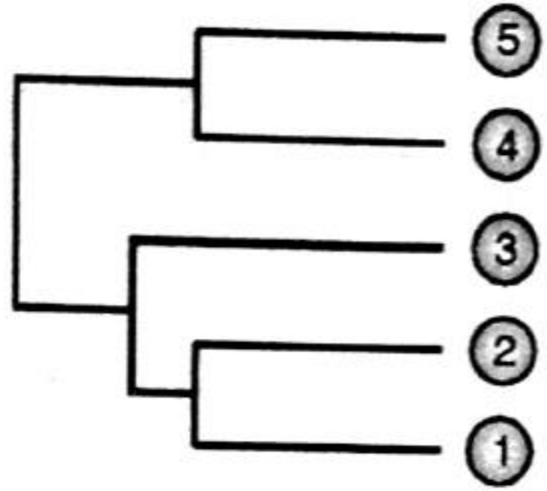
Step 3 :

$$\begin{array}{cc} & \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} & \begin{bmatrix} 0 & & & & \\ 2 & 0 & & & \\ 6 & 3 & 0 & & \\ 10 & 9 & 7 & 0 & \\ 9 & 8 & 5 & 4 & 0 \end{bmatrix} \end{array} \Rightarrow \begin{array}{cc} & \begin{array}{cccc} (1,2) & 3 & 4 & 5 \end{array} \\ \begin{array}{c} (1,2) \\ 3 \\ 4 \\ 5 \end{array} & \begin{bmatrix} 0 & & & \\ 3 & 0 & & \\ 9 & 7 & 0 & \\ 8 & 5 & 4 & 0 \end{bmatrix} \end{array} \Rightarrow \begin{array}{cc} & \begin{array}{ccc} (1,2,3) & 4 & 5 \end{array} \\ \begin{array}{c} (1,2,3) \\ 4 \\ 5 \end{array} & \begin{bmatrix} 0 & & \\ 7 & 0 & \\ 5 & 4 & 0 \end{bmatrix} \end{array}$$

$$\Downarrow$$

$$\begin{array}{cc} & \begin{array}{cc} (1,2,3) & 4,5 \end{array} \\ \begin{array}{c} (1,2,3) \\ (4,5) \end{array} & \begin{bmatrix} 0 & \\ 5 & 0 \end{bmatrix} \end{array}$$

$$d_{(1,2,3),(4,5)} = \min\{d_{(1,2,3),4}, d_{(1,2,3),5}\} = \min\{7, 5\} = 5$$



Complete Link :

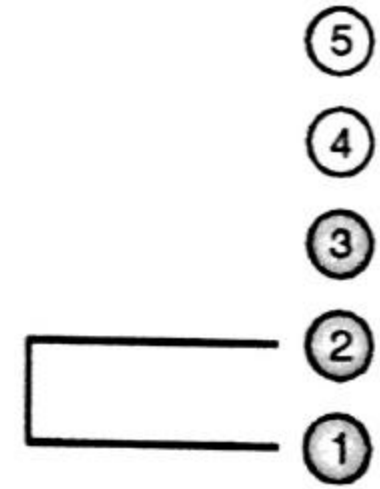
Step 1 :

$$\begin{array}{cc} & \begin{matrix}1 & 2 & 3 & 4 & 5\end{matrix} \\ \begin{matrix}1\\2\\3\\4\\5\end{matrix} & \begin{bmatrix}0 & & & & \\ 2 & 0 & & & \\ 6 & 3 & 0 & & \\ 10 & 9 & 7 & 0 & \\ 9 & 8 & 5 & 4 & 0\end{bmatrix}\end{array} \Rightarrow \begin{array}{cc} & \begin{matrix}(1,2) & 3 & 4 & 5\end{matrix} \\ \begin{matrix}(1,2)\\3\\4\\5\end{matrix} & \begin{bmatrix}0 & & & \\ 6 & 0 & & \\ 10 & 7 & 0 & \\ 9 & 5 & 4 & 0\end{bmatrix}\end{array}$$

$$d_{(1,2),3} = \max\{d_{1,3}, d_{2,3}\} = \max\{6, 3\} = 6$$

$$d_{(1,2)4} = \max\{d_{,1,4}, d_{2,4}\} = \max\{10, 9\} = 10$$

$$d_{(1,2)5} = \max\{d_{,1,5}, d_{2,5}\} = \max\{9, 8\} = 9$$

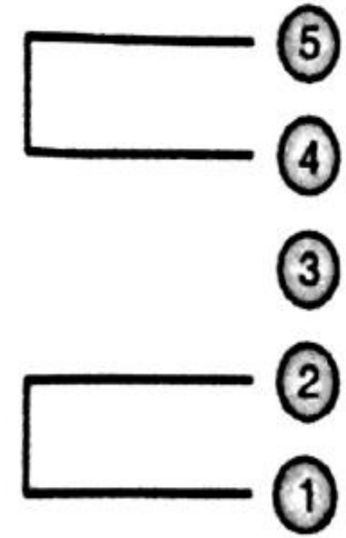


Step 2 :

$$\begin{array}{cc} & \begin{matrix}1 & 2 & 3 & 4 & 5\end{matrix} \\ \begin{matrix}1\\2\\3\\4\\5\end{matrix} & \begin{bmatrix}0 & & & & \\ 2 & 0 & & & \\ 6 & 3 & 0 & & \\ 10 & 9 & 7 & 0 & \\ 9 & 8 & 5 & 4 & 0\end{bmatrix}\end{array} \Rightarrow \begin{array}{cc} & \begin{matrix}(1,2) & 3 & 4 & 5\end{matrix} \\ \begin{matrix}(1,2)\\3\\4\\5\end{matrix} & \begin{bmatrix}0 & & & \\ 6 & 0 & & \\ 10 & 7 & 0 & \\ 9 & 5 & 4 & 0\end{bmatrix}\end{array} \Rightarrow \begin{array}{cc} & \begin{matrix}(1,2) & 3 & (4,5)\end{matrix} \\ \begin{matrix}(1,2)\\3\\(4,5)\end{matrix} & \begin{bmatrix}0 & & \\ 6 & 0 & \\ 10 & 7 & 0\end{bmatrix}\end{array}$$

$$d_{(1,2),(4,5)} = \max\{d_{(1,2),4}, d_{(1,2),5}\} = \max\{10, 9\} = 10$$

$$d_{3,(4,5)} = \max\{d_{3,4}, d_{3,5}\} = \max\{7, 5\} = 7$$



Step 3 :

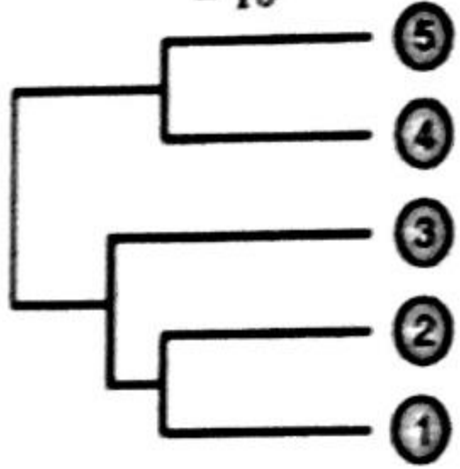
$$
\begin{array}{c|ccccc}
 & 1 & 2 & 3 & 4 & 5 \\
\hline
1 & 0 & & & & \\
2 & 2 & 0 & & & \\
3 & 6 & 3 & 0 & & \\
4 & 10 & 9 & 7 & 0 & \\
5 & 9 & 8 & 5 & 4 & 0
\end{array}
\Rightarrow
\begin{array}{c|cccc}
 & (1,2) & 3 & 4 & 5 \\
\hline
(1,2) & 0 & & & \\
3 & 6 & 0 & & \\
4 & 10 & 7 & 0 & \\
5 & 9 & 5 & 4 & 0
\end{array}
\Rightarrow
\begin{array}{c|ccc}
 & (1,2) & 3 & (4,5) \\
\hline
(1,2) & 0 & & \\
3 & 6 & 0 & \\
(4,5) & 10 & 7 & 0
\end{array}
$$

$$
\Downarrow
$$

$$
\begin{array}{c|cc}
 & (1,2,3) & (4,5) \\
\hline
(1,2,3) & 0 & \\
(4,5) & 10 & 0
\end{array}
$$

$$
d_{(1,2,3),(4,5)} = \max\{d_{(1,2),(4,5)}, d_{3(4,5)}\} = \max\{10, 7\}
$$

$$
= 10
$$



Average link :

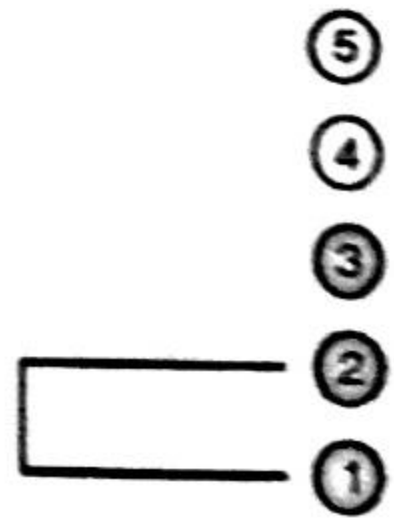
Step 1 :

$$\begin{array}{c}\\ 1\\ 2\\ 3\\ 4\\ 5\end{array}\begin{array}{c}\begin{array}{ccccc}1 & 2 & 3 & 4 & 5\end{array}\\ \begin{bmatrix}0 & & & & \\ 2 & 0 & & & \\ 6 & 3 & 0 & & \\ 10 & 9 & 7 & 0 & \\ 9 & 8 & 5 & 4 & 0\end{bmatrix}\end{array} \Rightarrow \begin{array}{c}\\ (1,2)\\ 3\\ 4\\ 5\end{array}\begin{array}{c}\begin{array}{cccc}(1,2) & 3 & 4 & 5\end{array}\\ \begin{bmatrix}0 & & & \\ 4.5 & 0 & & \\ 9.5 & 7 & 0 & \\ 8.5 & 5 & 4 & 0\end{bmatrix}\end{array}$$

$$d_{(1,2),3} = \frac{1}{2}(d_{1,3} + d_{2,3}) = \frac{6+3}{2} = 4.5$$

$$d_{(1,2),4} = \frac{1}{2}(d_{1,4} + d_{2,4}) = \frac{10+9}{2} = 9.5$$

$$d_{(1,2),5} = \frac{1}{2}(d_{1,5} + d_{2,5}) = \frac{9+8}{2} = 8.5$$



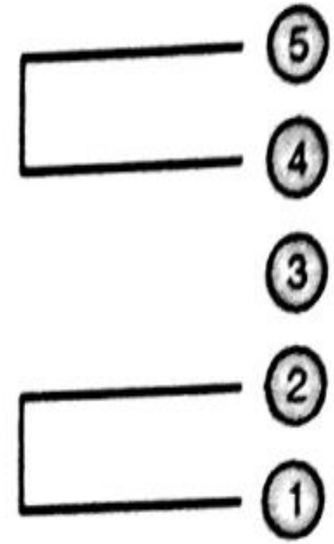
Step 2 :

$$
\begin{array}{cc}
 & \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\
\begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} &
\begin{bmatrix} 0 & & & & \\ 2 & 0 & & & \\ 6 & 3 & 0 & & \\ 10 & 9 & 7 & 0 & \\ 9 & 8 & 5 & 4 & 0 \end{bmatrix}
\end{array}
\Rightarrow
\begin{array}{cc}
 & \begin{array}{cccc} (1, 2) & 3 & 4 & 5 \end{array} \\
\begin{array}{c} (1, 2) \\ 3 \\ 4 \\ 5 \end{array} &
\begin{bmatrix} 0 & & & \\ 4.5 & 0 & & \\ 9.5 & 7 & 0 & \\ 8.5 & 5 & 4 & 0 \end{bmatrix}
\end{array}
\Rightarrow
\begin{array}{cc}
 & \begin{array}{ccc} (1,2) & 3 & (4,5) \end{array} \\
\begin{array}{c} (1,2) \\ 3 \\ (4,5) \end{array} &
\begin{bmatrix} 0 & & \\ 4.5 & 0 & \\ 9 & 6 & 0 \end{bmatrix}
\end{array}
$$

$$d_{(1,2),(4,5)} = \frac{1}{4}\{d_{1,4,} + d_{1,5} + d_{2,4} + d_{2,5}\} = 1/4\ \{\ 10 + 9 + 9 + 8\}$$

$$= 9$$

$$d_{3,(4,5)} = \frac{1}{2}\{d_{3,4,} + d_{3,5}\} = 6$$



Step 3 :

$$
\begin{array}{c}
 \\
\begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array}
\end{array}
\begin{array}{c}
\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\
\begin{bmatrix} 0 & & & & \\ 2 & 0 & & & \\ 6 & 3 & 0 & & \\ 10 & 9 & 7 & 0 & \\ 9 & 8 & 5 & 4 & 0 \end{bmatrix}
\end{array}
\Rightarrow
\begin{array}{c}
 \\
\begin{array}{c} (1,2) \\ 3 \\ 4 \\ 5 \end{array}
\end{array}
\begin{array}{c}
\begin{array}{cccc} (1,2) & 3 & 4 & 5 \end{array} \\
\begin{bmatrix} 0 & & & \\ 4.5 & 0 & & \\ 9.5 & 7 & 0 & \\ 8.5 & 5 & 4 & 0 \end{bmatrix}
\end{array}
\Rightarrow
\begin{array}{c}
 \\
\begin{array}{c} (1,2) \\ 3 \\ (4,5) \end{array}
\end{array}
\begin{array}{c}
\begin{array}{ccc} (1,2) & 3 & (4,5) \end{array} \\
\begin{bmatrix} 0 & & \\ 4.5 & 0 & \\ 9 & 6 & 0 \end{bmatrix}
\end{array}
$$

$$\Downarrow$$

$$
\begin{array}{c}
 \\
\begin{array}{c} (1,2,3) \\ (4,5) \end{array}
\end{array}
\begin{array}{c}
\begin{array}{cc} (1,2,3) & (4,5) \end{array} \\
\begin{bmatrix} 0 & \\ 8 & 0 \end{bmatrix}
\end{array}
$$

$$d_{(1,2,3),(4,5)} = \frac{1}{6}(d_{1,4} + d_{1,5} + d_{2,4} + d_{2,5} + d_{3,4} + d_{3,5}) = 1/6\,\{10 + 9 + 9 + 8 + 7 + 5\}$$

$$= 8$$